



Engineering Exploration (A4022) Course End Project (CEP) 2019-20

**Sanitation in urban Settings
Team#02**



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Outline of The Presentation

- Introduction
- Objectives of the project
- Block Diagram / Circuit Diagram
- Project Description
- Features , Advantages , & Applications
- Conclusions and Future Extensions
- References



Introduction

Drainage Water level Sensor

In this project we have developed a water level sensor to detect the over flow of drainage . Most of the cities have been suffering from over flow of drainage and this is causing a lot of diseases spread through a over flow drainage system , Mainly During Rainy seasons.





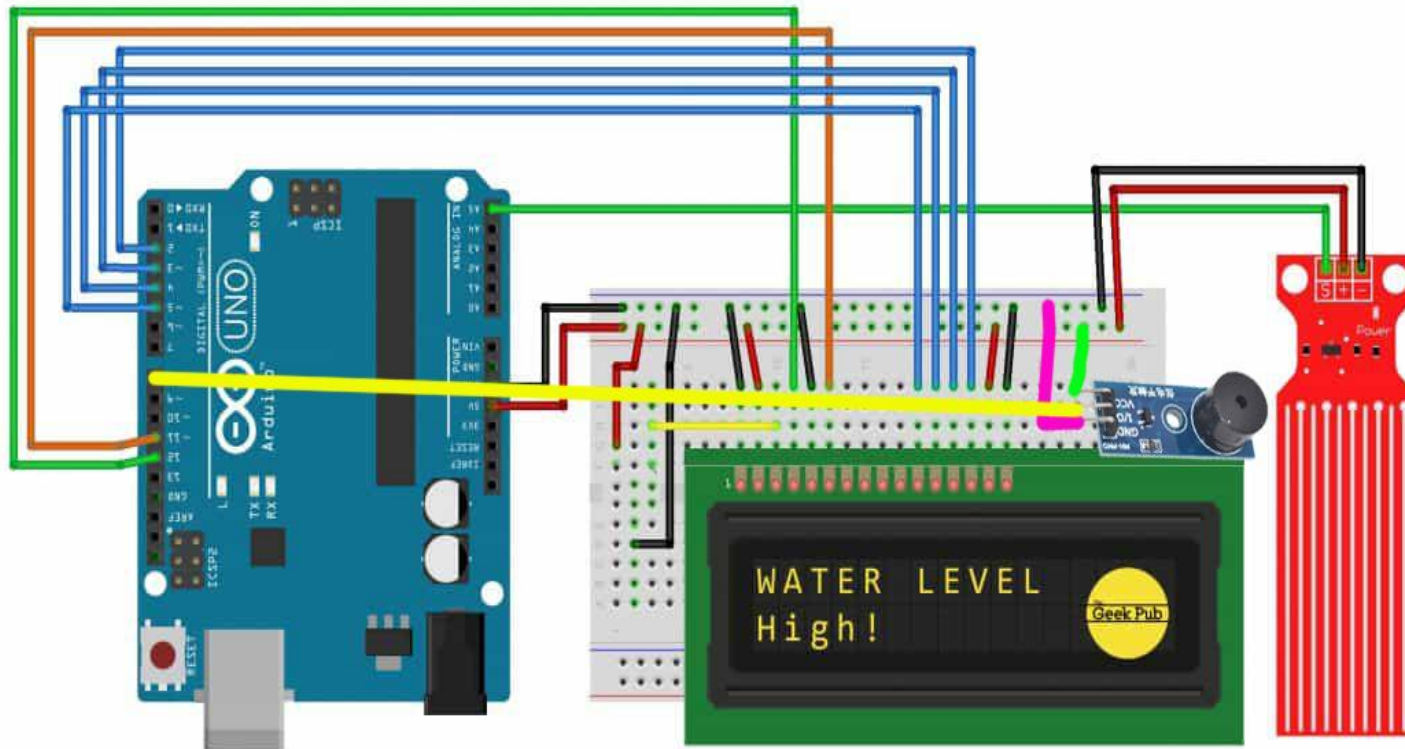
Objectives of the project

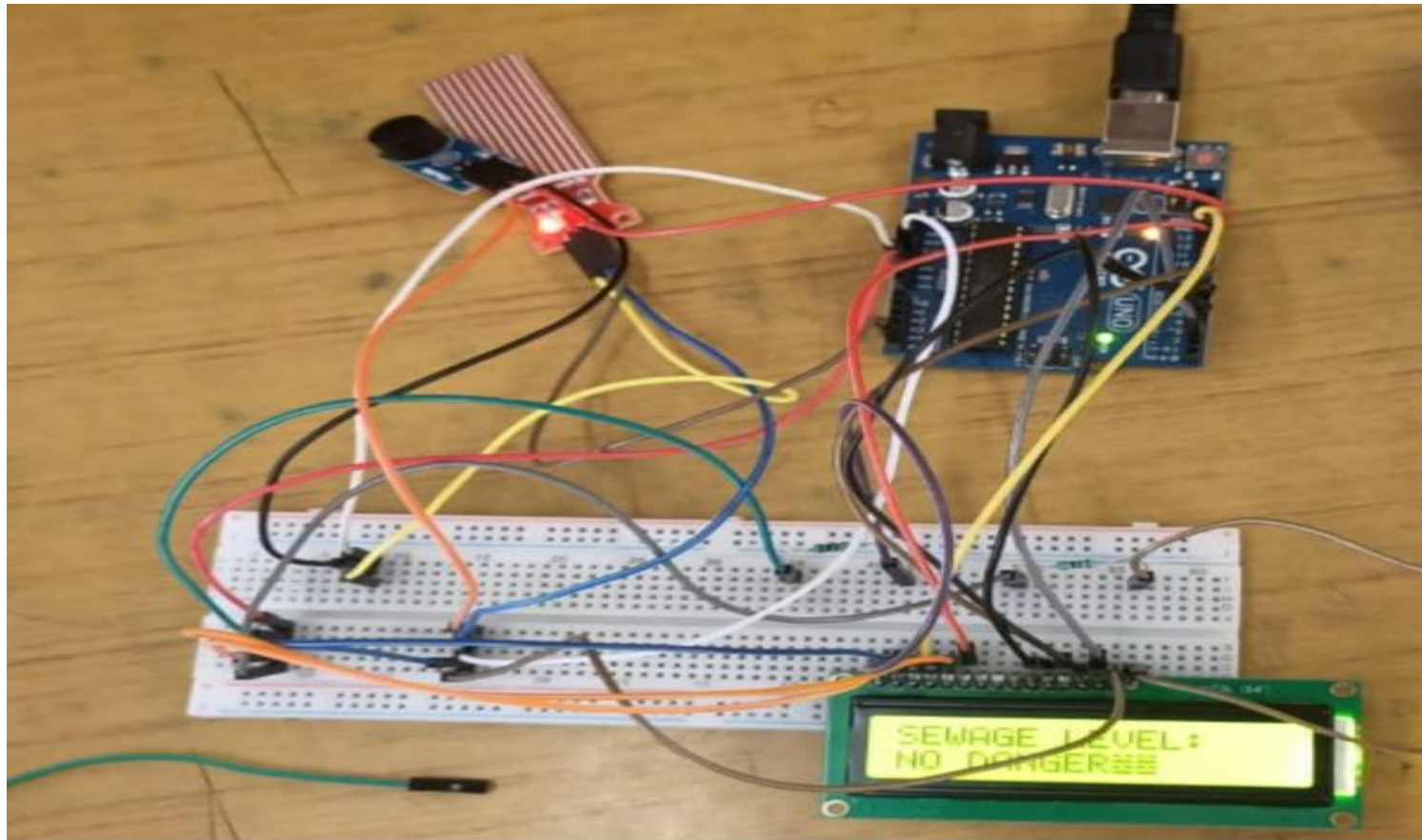
- The Main objective of the Project is to maintain a Clean City .
- Clean the Drainage system before it's full and over flown.
- Maintain a proper regular flow.





Block Diagram / Circuit Diagram







Project Description

Components

- Arduino Board Uno
- Bread board
- Water Sensor
- Lcd Display
- Buzzer
- Resistors



The water sensor is kept below the Manhole or drainage system ,As the Water increases at certain extend it has to Touch the water sensor.

As soon as the Drainage water over flows and touches the water sensor ,The water level is displaced in Lcd Display and further flow of water the Level is indicated and Buzzer gets On giving a alert of over flow through man hole.





Features , Advantages , & Applications

- Over Flow of the Man holes and drainage systems can be Detected.
- It can cover the whole city and can show the Man holes which are going to be over Flown
- This can used for also used in large tanks at various Industrial Purposes from the fluids getting Over flown from the chemical tanks in Industries.



Conclusions and Future Extensions

- This Device is less expensive and can be operated at the center.
- Multiple manholes can be watched and alerted at a time.
- For further future Extensions we can use this device for checking of flow of Drainage system



References

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- D.E Hesmondhalghand E.R Laithwaite ,method of analyzing properties of 2-phaseservomotor and AC tachometer,Volume70



Thank You!!!

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